Authors

Y. Salinas-Reyes[1,2], Y. Chen[3], T.M. Vaughan[3], Hill, N.J.,[3]J.S. Carp[3], E. Meyers[1,2]

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[1]Bioengineering, Univ. of Texas at Dallas, Richardson, TX; [2]Texas Biomed. Device Ctr; [3]Natl. Ctr. for Adaptive Neurotechnologies, Stratton VA Med. Ctr., Albany, NY

Disclosures

Y. Salinas-Reyes: None; Y. Chen: None; T.M. Vaughan: None; N.J. Hill: None; J.S. Carp: None; E. Meyers: None.

Title: Closed-loop vagus nerve stimulation as a reinforcement signal for H-reflex operant conditioning

Abstract:

Spasticity after stroke and spinal cord injury is a major driver of disability, yet effective treatments that selectively target spinal reflex circuits underlying spasticity remain limited. H-reflex (HR) operant conditioning (HROC) reduces reflex excitability by delivering a reinforcement when HR amplitude falls below criterion. Pre-clinical studies demonstrate successful reflex modulation, and early clinical trials suggest functional improvement. However, HROC's reliance on explicit, attention-demanding reinforcement over prolonged training timelines (8 weeks of 24/7 conditioning pre-clinically; 1-hour sessions, 3x week for 12 weeks clinically) may exclude individuals unable to sustain prolonged cognitive engagement and limit translational scalability. Vagus nerve stimulation (VNS), delivered closed-loop with training events, engages locus coeruleus-noradrenergic and raphe-serotonergic projections to the spinal cord, delivering a timed neuromodulatory signal to the spinal circuits underlying HR amplitude — the same circuits targeted by conventional food-reinforced HROC. In our V-HROC paradigm, successful down-conditioning trials (defined as HR amplitude falling below the 30th percentile of each rat's HR size distribution) receive VNS reinforcement. We hypothesize that VNS delivered following successful conditioning trials can substitute for food-pellet reward reinforcement by serving as a neuromodulatory reinforcement signal, producing HR down-conditioning comparable to food-reinforced HROC. Adult female Sprague-Dawley rats chronically implanted with vagus and tibial nerve stimulating cuffs undergo HROC, in which HRs are elicited via tibial nerve stimulation and soleus responses are captured from bipolar implanted EMG electrodes. Down-conditioning trials determined to be successful receive either: (1) VNS; (2) standard food pellet reward; or (3) no reinforcement (no VNS or food-pellet). This design allows us to assess how reinforcement modality shapes HR plasticity; the no-reward group confirms that HR changes in the CNV group are driven by VNS acting as a reward, not by VNS stimulation alone suppressing the reflex. Primary outcomes are

rate of down-conditioning (time to reach target reduction) and magnitude of suppression (total reduction in reflex amplitude); group comparisons assess whether VNS can substitute for food reinforcement in driving HR plasticity. These experiments test whether VNS-mediated neuromodulatory reinforcement is sufficient to drive spinal operant plasticity, providing a mechanistic foundation for attention-independent HROC protocols targeting hyperreflexia and spasticity.

Potential Key Words:

Current Selected (does order matter here?)

Neuromodulation

Rehabilitation

Electrophysiology

Other candidates

Reinforcement Learning

Injury

Spinal Cord Injury

Stroke

EMG?

Instrumental learning?

Potential Themes:

Current Selected

J.08.a. Methods to Modulate Neural Activity – Electrical

F.06.b. Posture and gait – Reflexes and low-level control

Other candidates

H.02.c. Reward and appetitive learning and memory – Neural mechanisms

D.05.g. Spinal cord injury – Training, rehabilitation, and repair

D.03.d. Stroke recovery – Non-pharmacological approaches to therapy

H.02.a. Reward and appetitive learning and memory – Acquisition and memory modification

H.02.b. Reward and appetitive learning and memory – Neural circuits

F.09.a. Motor neurons – Activity, sensory, and central control – Exercise, injury, and disease

D.05.h. Spinal cord injury – Plasticity and recovery

D.03.b. Stroke, damage, or disease – Mechanisms of abnormal function

Funding Sources:

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